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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 4 – Réseaux I

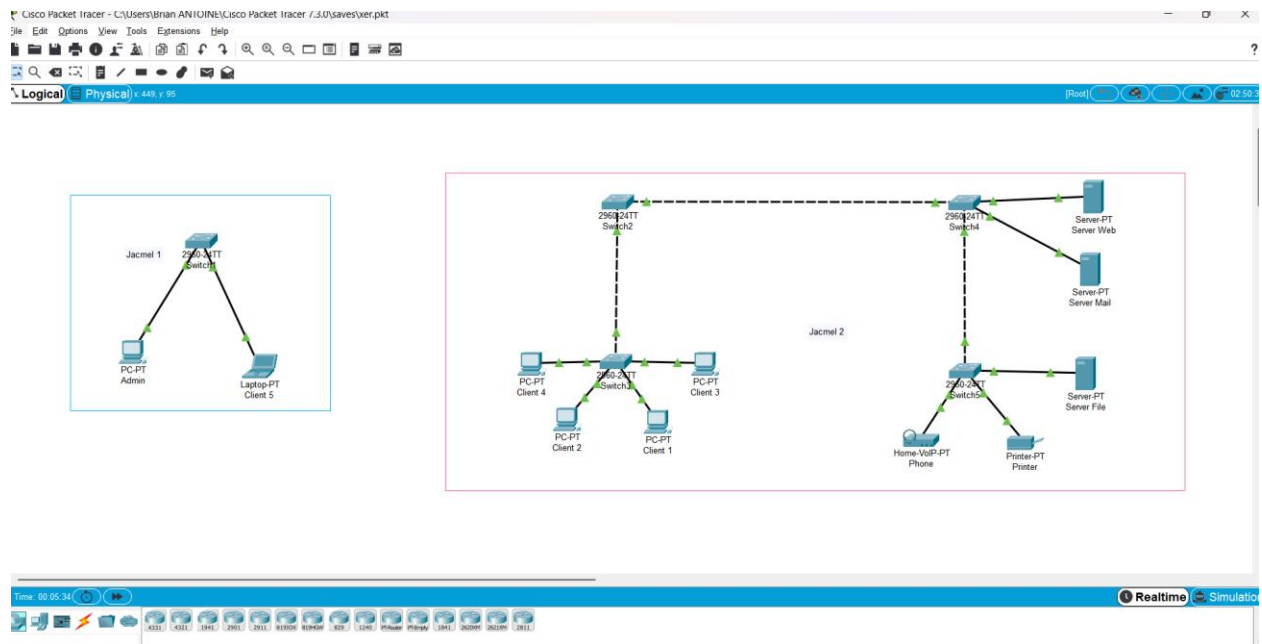
Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

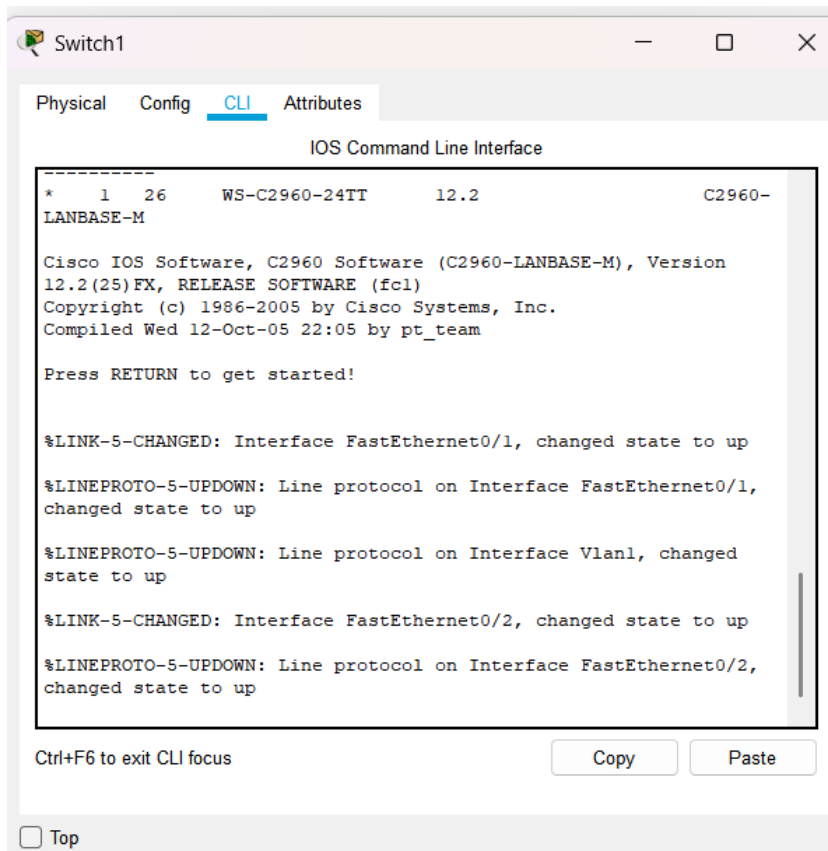
Niveau : L3

Date: 22/11/2025

➤ Reproduction de la topologie



➤ Configuration des Switchs



Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
-----
* 1 26 WS-C2960-24TT 12.2 C2960-
LANBASE-M

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version
12.2(25)FX, RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up
```

Ctrl+F6 to exit CLI focus

Copy Paste

☐ Top

Switch3

Physical Config CLI Attributes

IOS Command Line Interface

```
changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5,
changed state to up
```

Ctrl+F6 to exit CLI focus

Copy Paste

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Switch4

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
```

Ctrl+F6 to exit CLI focus

CopyPaste

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Switch5

Physical Config CLI Attributes

IOS Command Line Interface

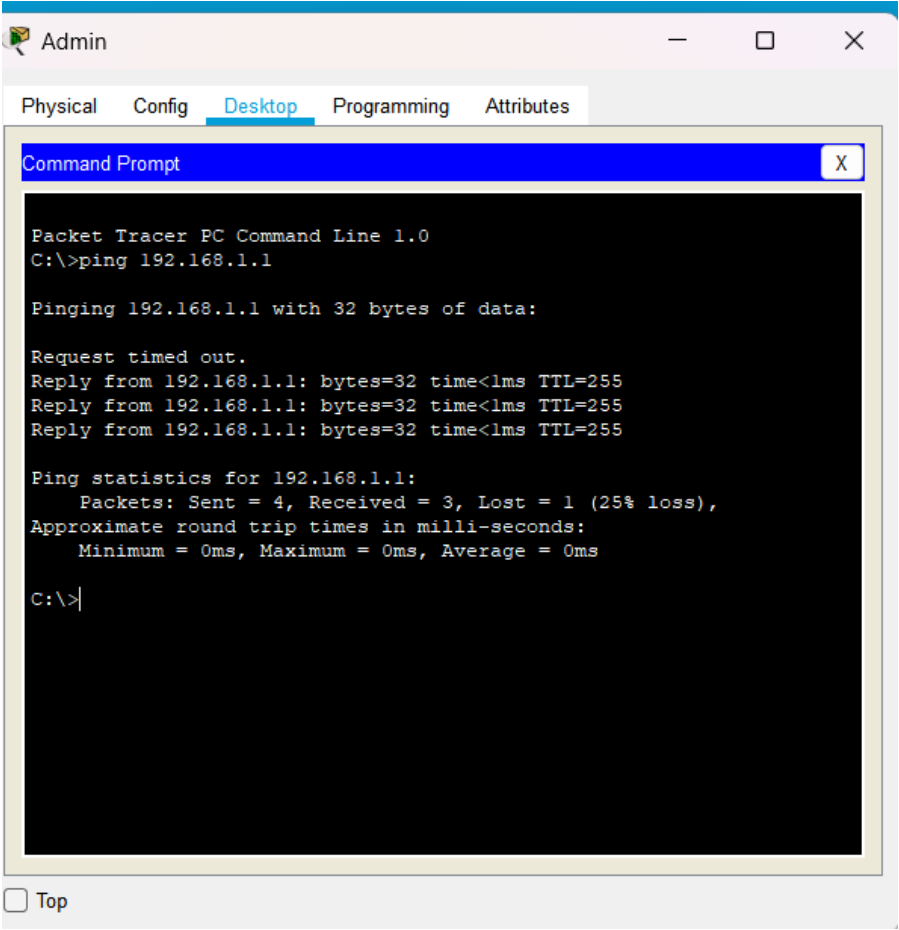
```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4,
changed state to up
```

Ctrl+F6 to exit CLI focus

CopyPaste

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➤ Test de connectivité



The screenshot shows a Packet Tracer PC Command Line window titled "Admin". The "Desktop" tab is selected. Inside the window is a "Command Prompt" titled "Command Prompt" with a close button "X". The command prompt shows the following text:

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

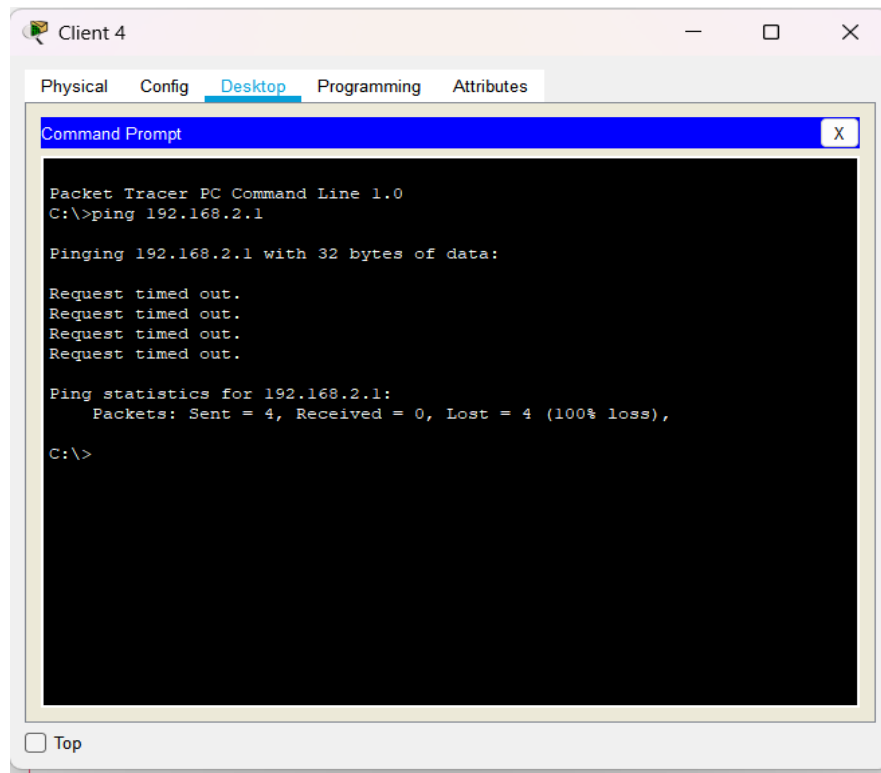
Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

At the bottom left of the window, there is a checkbox labeled "Top".

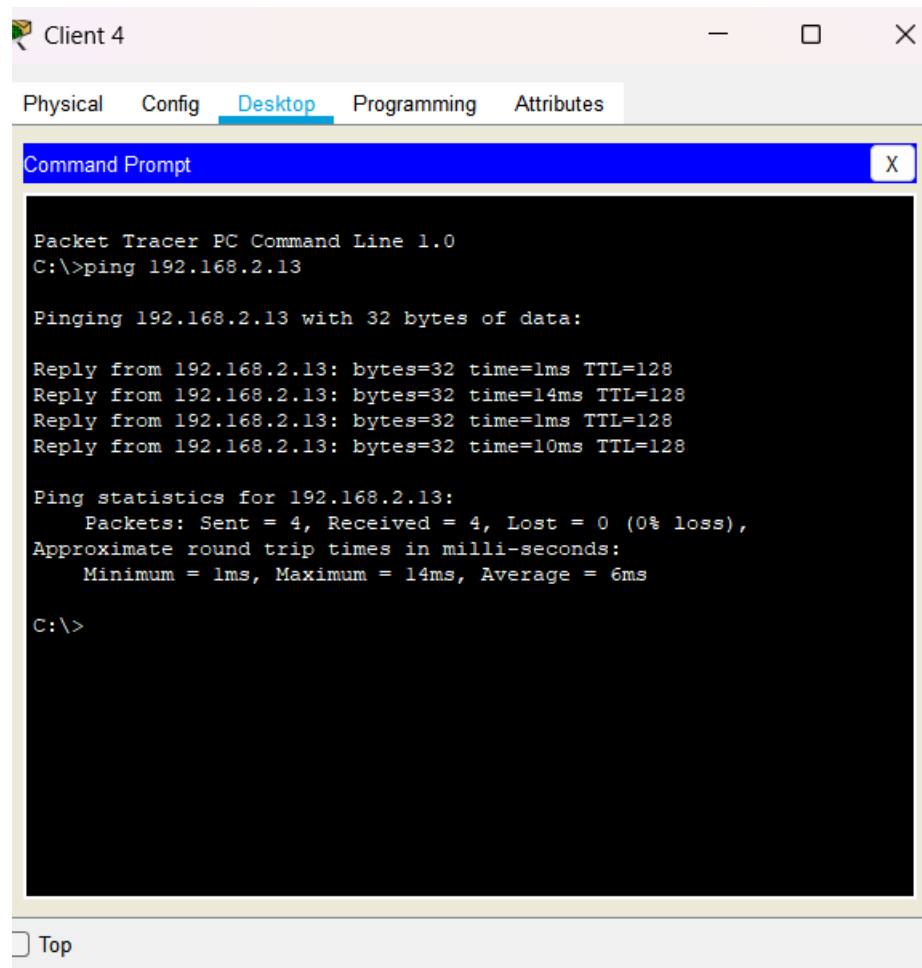


✓ Simulation

Realtime Simulation											
	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
		Failed	Switch2	Switch1	ICMP		6.830	N	1	(edit)	(delete)
		Failed	Switch2	Switch3	ICMP		7.873	N	2	(edit)	(delete)
		Failed	Switch4	Switch5	ICMP		7.876	N	3	(edit)	(delete)

1. Reproduisez cette topologie en configurant le routeur et les switches, puis en attribuant les adresses IP aux dispositifs. Utilisez soit IPv4, soit IPv6, et testez la connectivité des deux VLAN à l'aide de la commande ping et du mode de simulation.

▪ Reproduction de la topologie



✓ Test de Simulation

The screenshot shows the Realtime Simulation tab in Packet Tracer. A table displays network events, including successful pings from various sources to destinations.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Switch1	Admin	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Switch4	Switch5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Switch3	Client 1	ICMP		0.000	N	3	(edit)	(delete)

